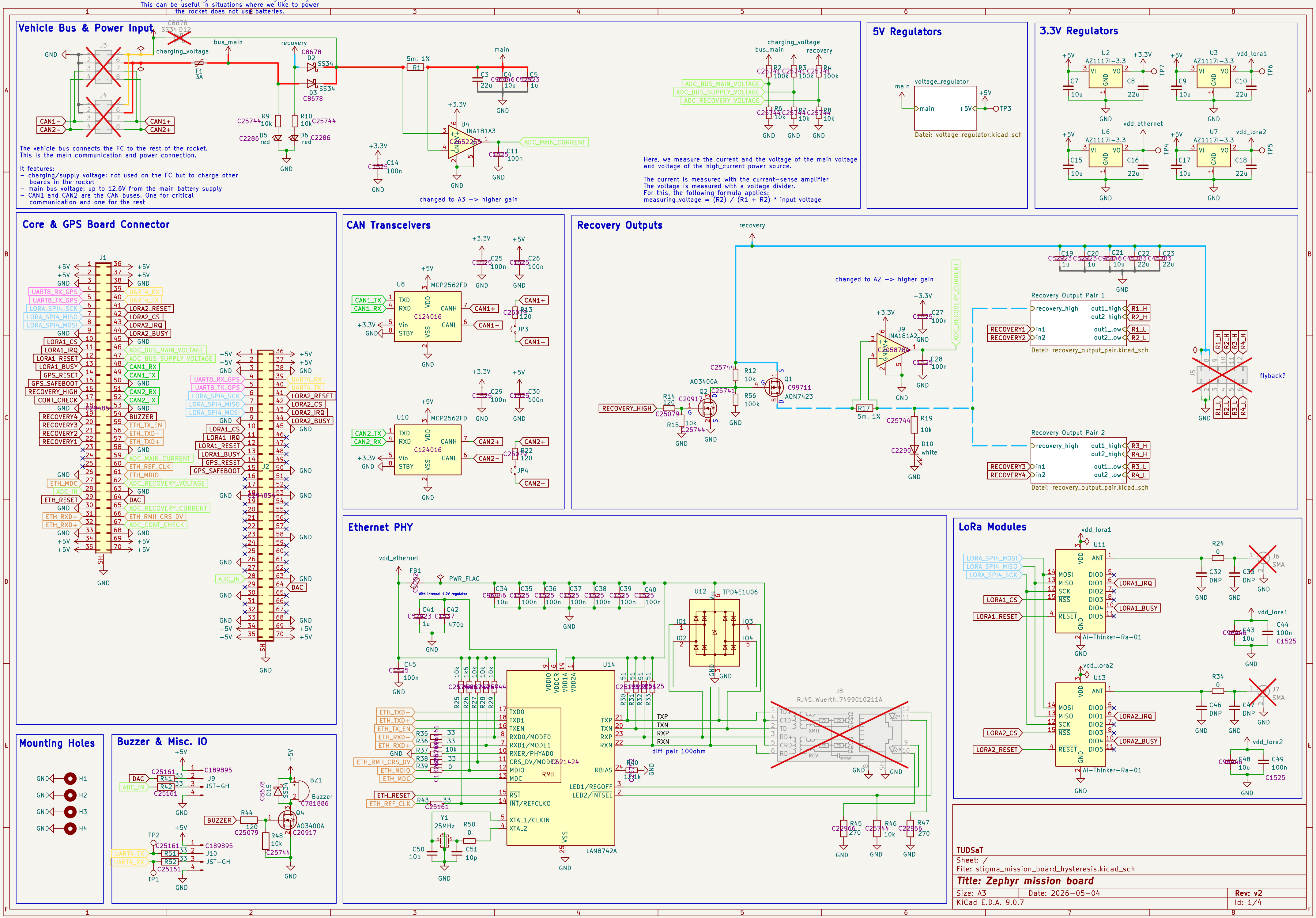
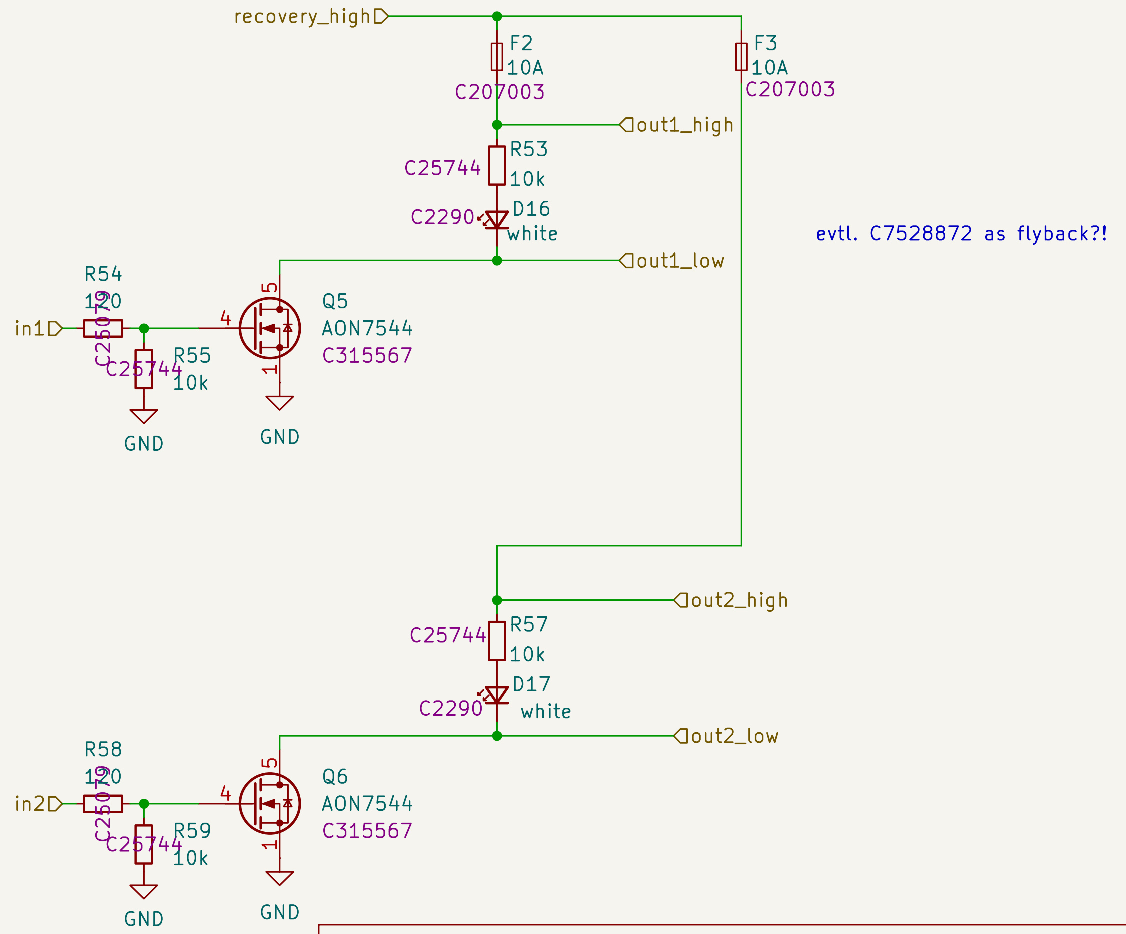

D12 allows powering the FC via the charging voltage.
This can be useful in situations where we like to power
the rocket does not use batteries.
Vehicle Bus & Power Input
The vehicle bus connects the FC to the rest of the rocket.
This is the main communication and power connection.
It features:
- charging/supply voltage: not used on the FC but to charge other boards in the rocket
- main bus voltage: up to 12.6V from the main battery supply
- CAN1 and CAN2 are the CAN buses. One for critical communication and one for the rest
changed to A3 -> higher gain
Here, we measure the current and the voltage of the main voltage and voltage of the high_current power source.
The current is measured with the current-sense amplifier
The voltage is measured with a voltage divider.
For this, the following formula applies:
measuring_voltage = (R2) / (R1 + R2) * input voltage
5V Regulators
voltage_regulator
Datei: voltage_regulator.kicad_sch
3.3V Regulators
Core & GPS Board Connector
CAN Transceivers
Recovery Outputs
changed to A2 -> higher gain
Recovery Output Pair 1
Datei: recovery_output_pair.kicad_sch
Recovery Output Pair 2
Datei: recovery_output_pair.kicad_sch
flyback?
Ethernet PHY
diff pair 100ohm
LoRa Modules
Mounting Holes
Buzzer & Misc. IO
TUDSaT
Sheet: /
File: stigma_mission_board_hysteresis.kicad_sch
Title: Zephyr mission board
Size: A3
Date: 2026-05-04
Rev: v2
KiCad E.D.A. 9.0.7
Id: 1/4

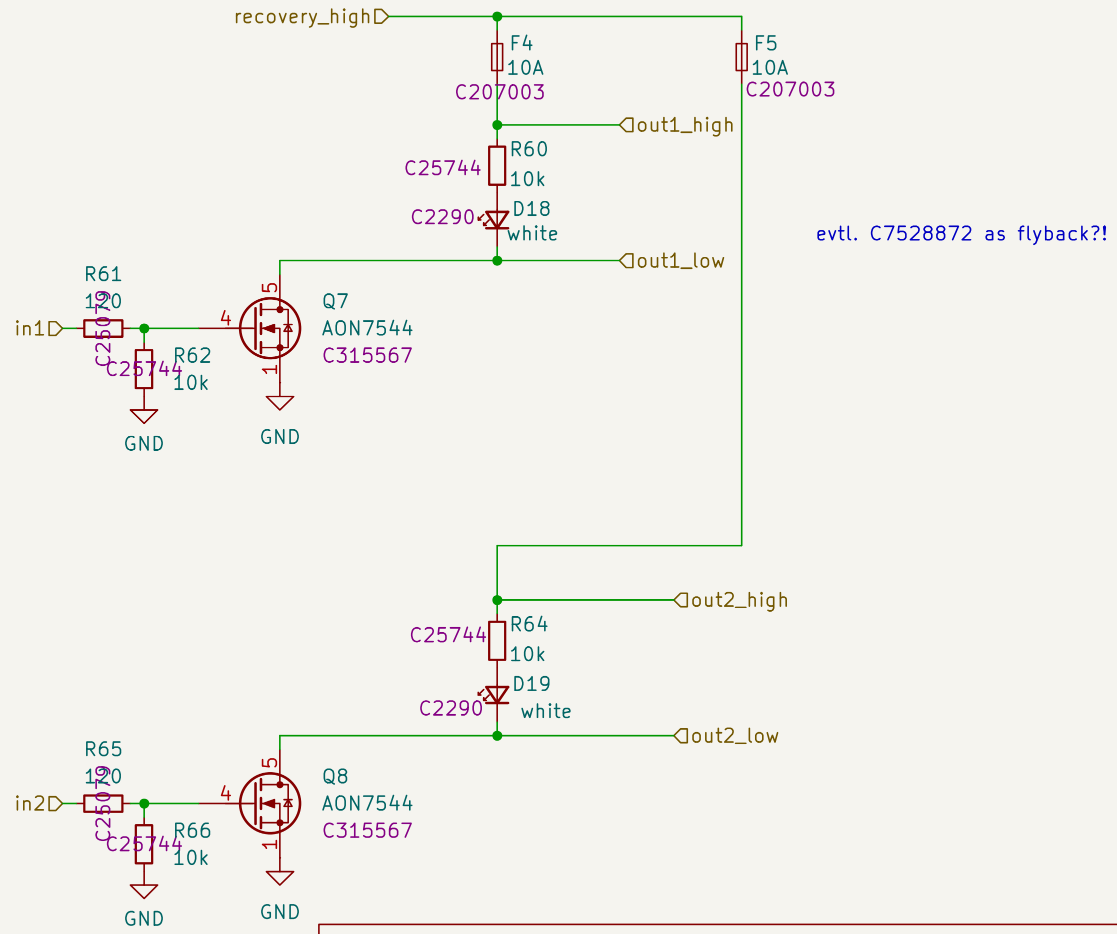


Sheet: /Recovery Output Pair 1/
File: recovery_output_pair.kicad_sch

Title:

Size: A5 | Date: | **Rev:**

KiCad E.D.A. 9.0.7 | Id: 2/4



Sheet: /Recovery Output Pair 2/

File: recovery_output_pair.kicad_sch

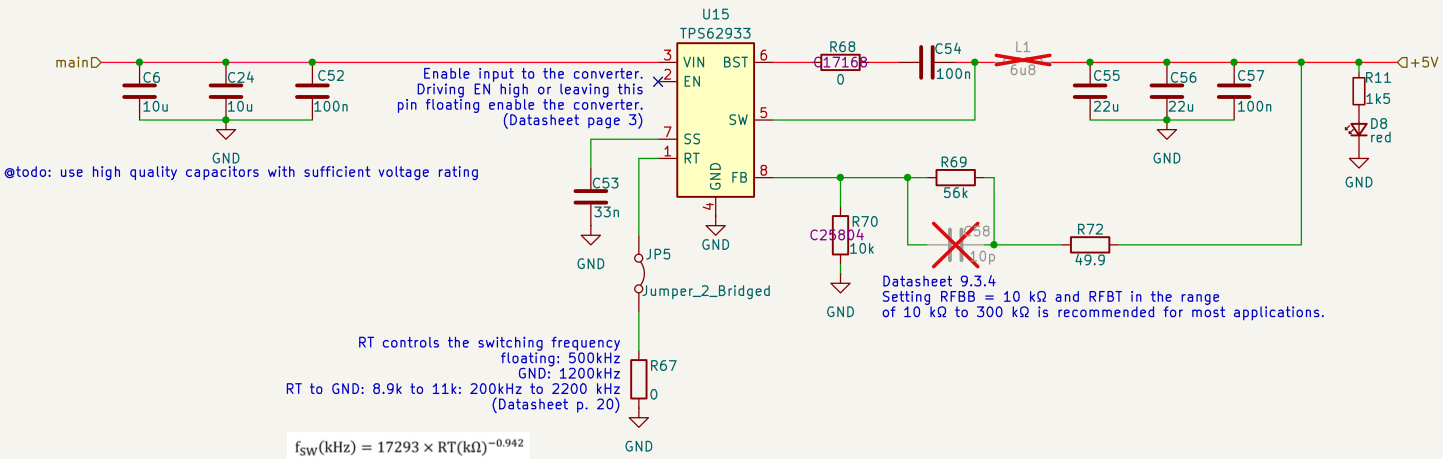
Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 3/4

10.2.2.5 Bootstrap Capacitor Selection

A 0.1-µF ceramic capacitor must be connected between the BST to SW pins for proper operation. TI recommends to use a ceramic capacitor with X5R or better grade dielectric. The capacitor C5 must have a 16-V or higher voltage rating.

In addition, adding one BST resistor R4 to reduce the spike voltage on the SW node, TI recommends the resistance smaller than 10 Ω be used between BST to the bootstrap capacitor.



$$f_{SW}(\mathrm{kHz}) = 17293 \times RT(\mathrm{k\Omega})^{-0.942}$$

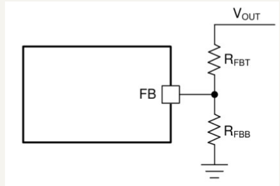


Figure 9-2. Output Voltage Setting

$$R_{FBT} = \frac{V_{OUT} - V_{REF}}{V_{REF}} \times R_{FBB}$$

where

- V_{REF} is the 0.8 V (the internal reference voltage).
- R_{FBB} is 10 kΩ (recommended).

Sheet: /voltage_regulator/
File: voltage_regulator.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 4/4